

A.R.T.E. – BOM (Bill of Materials)

Version A (Low-Cost Robust)

Version B (High-Performance Upgrade)

Modularly designed according to system components.

1. METAL-FUEL REACTOR (Energy Unit)

Common Base Components

Position	Component	Specification	Quantity
1	Reactor vessel	Stainless steel, 3–4 mm, Ø 28 cm, H 45 cm	1
2	Filler opening	12–15 cm, screw cap	1
3	Waste container	2–3 L, Stainless steel/PE	1
4	Heat exchanger	Stainless steel spiral	1
5	Temperature sensor	NTC/PT100	2
6	Pressure sensor	0–6 bar	1
7	Pressure relief valve	3–4 bar	1
8	Water reservoir	10–15 L	1
9	Water pipe	EPDM/PU	1 Set
10	Seals	High-temperature	1 Set

Version A – Low-Cost

Position	Component	Specification	Quantity
A1	Thermoelectric Generators (TEG)	200–400 W total	4–6 Modules
A2	Passive heat sink	Aluminum	1
A3	Fan	48V	1
A4	Mechanical water supply valve	Float-controlled	1
A5	48V DC boost converter	1 kW	1

Version B – High-Performance

Position	Component	Specification	Quantity
B1	Micro steam turbine OR PEM fuel cell	1–2 kW	1
B2	Electronic water supply valve	PWM-controlled	1
B3	Liquid cooling	Pump + Radiator	1 Set
B4	Digital sensors	Pressure, Temp, Gas flow	1 Set
B5	Power Management (MPPT)	48V	1

2. A.R.T.E. FORGE – PELLET PRESS

2.1 Mechanical Base Structure

Position	Component	Specification	Quantity
1	Frame	Steel profiles 40×40×3 mm	1 Set
2	Flat die	Ø 400 mm, 25–30 mm thick	1
3	Rollers	Ø 120–150 mm, knurled	2
4	Roller bearings	Deep groove ball bearings	4
5	Gearbox	1:20	1
6	Main motor	48V, 1–2 kW	1
7	Hopper	Steel/PE	1
8	Agitator	48V Motor	1
9	Protective grid	12–15 mm	1
10	Drying hood	Sheet metal	1

2.2 Version A – Low-Cost

Position	Component	Specification	Quantity
A1	Fixed speed control	Standard ESC	1
A2	Mechanical backstop	Ratchet/Freewheel	1
A3	Passive warm air duct	Reactor waste heat	1

2.3 Version B – High-Performance

Position	Component	Specification	Quantity
B1	Speed sensor	Hall sensor	1
B2	Humidity sensor	IR/Capacitive	1
B3	Air blower	48V	1
B4	Electronic air control	PWM	1
B5	Temperature sensor drying	PT100	1

3. VEHICLE PLATFORM A.R.T.E. 2.0

3.1 Base Structure

Position	Component	Specification	Quantity
1	Frame	Steel profiles 40×40×3 mm	1 Set
2	Hub motors	48V, 500–800 W	4
3	Motor controller	48V	4
4	Tires	20–25 cm width	4
5	Battery	48V, 20–40 Ah	1
6	Electronics box	IP65	1
7	Wiring harness	48V	1 Set
8	Sensor mast	Steel/Alu	1

3.2 Version A – Low-Cost

Position	Component	Specification	Quantity
A1	Rigid axles	Steel	2
A2	Simple spindle for Terra-Engine	48V	1
A3	Basic battery	48V, 20 Ah	1

3.3 Version B – High-Performance

Position	Component	Specification	Quantity
B1	Partially suspended suspension	Rubber/Leaf spring	4
B2	FEM-optimized frame	CNC-cut	1
B3	High-performance battery	48V, 40–60 Ah	1

4. TERRA-ENGINE (Soil Healing)

4.1 Mechanics

Position	Component	Specification	Quantity
1	Support beam	60×40×4 mm Steel	1
2	Spikes	HRC 55–60, 30–40 cm	3
3	Spike flanges	Steel	3
4	Vibration motors	100–300 W	1–2
5	Rubber buffers	Vibration decoupling	4

4.2 Injection

Position	Component	Specification	Quantity
6	Pump	48V, 2–4 bar	1
7	Lines	EPDM/PU	1 Set
8	Filter	100–200 µm	1–2
9	Nozzles	1–2 mm	3
10	Tank	150–250 L	1

4.3 Version A – Low-Cost

Position	Component	Specification	Quantity
A1	Solenoid valve	48V	1
A2	Fixed flow	Nozzle size defined	–

4.4 Version B – High-Performance

Position	Component	Specification	Quantity
B1	Flow meter	0–5 L/min	1
B2	Pressure sensor	0–6 bar	1
B3	Regulated pump	PWM	1

5. AUTONOMY & SENSORS

5.1 Basic Sensors

Position	Component	Specification	Quantity
1	GPS	Standard GNSS	1
2	IMU	6–9 Axes	1
3	Lidar	2D, 10–30 m	1
4	Camera	720–1080p	1
5	Gore-Tex membrane	Dust protection	1 Set

5.2 Version A – Low-Cost

Position	Component	Specification	Quantity
A1	ESP32/STM32	Control	2–3
A2	Basic Lidar	2D	1
A3	Camera wiper	Mini-wiper	1

5.3 Version B – High-Performance

Position	Component	Specification	Quantity
B1	NVIDIA Jetson	Nano/Orin	1
B2	GPS-RTK	Rover + Base	1 Set
B3	3D-Lidar	360°	1
B4	Camera cleaning	Air blast	1

6. SOFTWARE

6.1 Version A – Low-Cost

Position	Software Module
A1	Motor control
A2	GPS navigation
A3	Lidar stop
A4	Terra-Engine ON/OFF
A5	Reactor monitoring
A6	Safety logic

6.2 Version B – High-Performance

Position	Software Module
B1	ROS2-Middleware
B2	Sensor fusion (GPS-RTK + IMU + Lidar)
B3	Mapping
B4	Mission Manager
B5	Telemetry Dashboard
B6	AI Modules (optional)

7. ACCESSORIES & CONSUMABLES

Position	Component	Quantity
1	Screws M6–M20	1 Set
2	Cable ties	1 Set
3	Hydraulic/EPDM seals	1 Set
4	Lubricating grease	1
5	Filter inserts	5–10
6	Replacement spikes	3
7	Replacement nozzles	3
8	Aluminum scrap	5–8 kg per filling
9	Water	6–10 L per filling